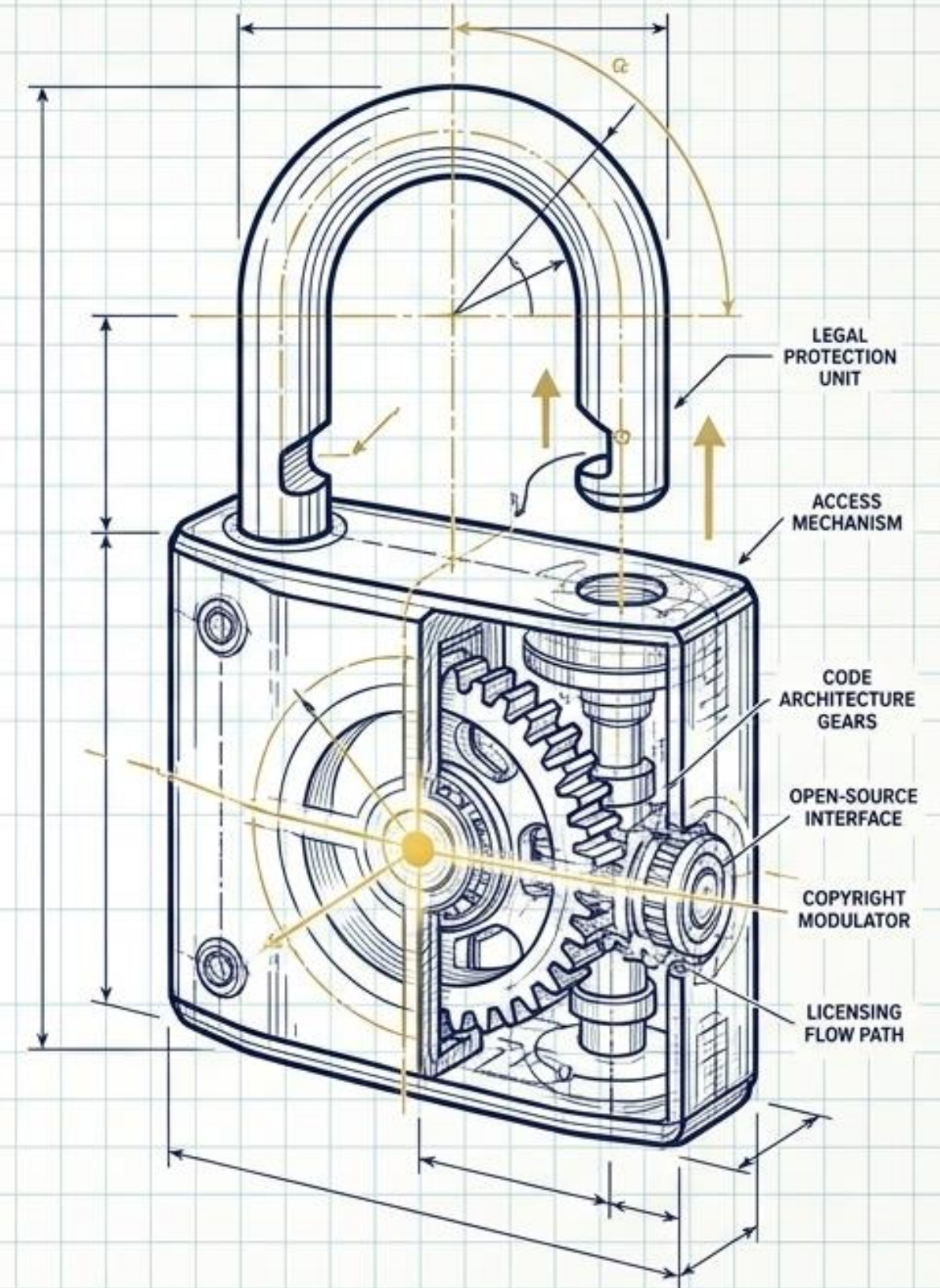


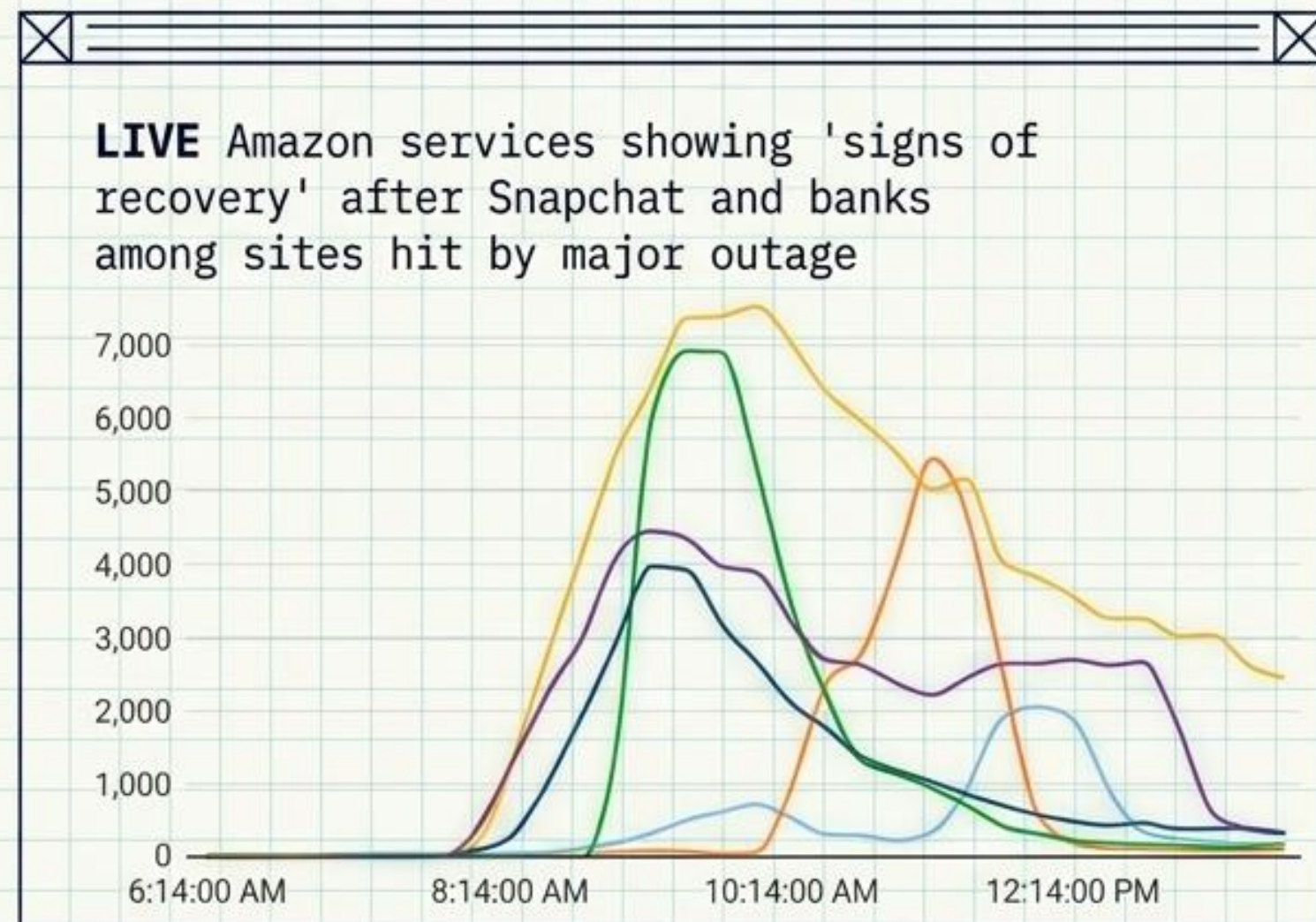
Scalable Advanced Software Solutions: Open-Source Licences and Copyright

A Developer's Guide to the
Legal Architecture of Code



When The Infrastructure Fails, Who Holds the Keys?

The Incident



The Reality

Software is omnipresent. It drives global finance, communications, and infrastructure.

But underneath the technology is a rigid legal framework dictating exactly who owns, controls, and can modify the code that runs the world.

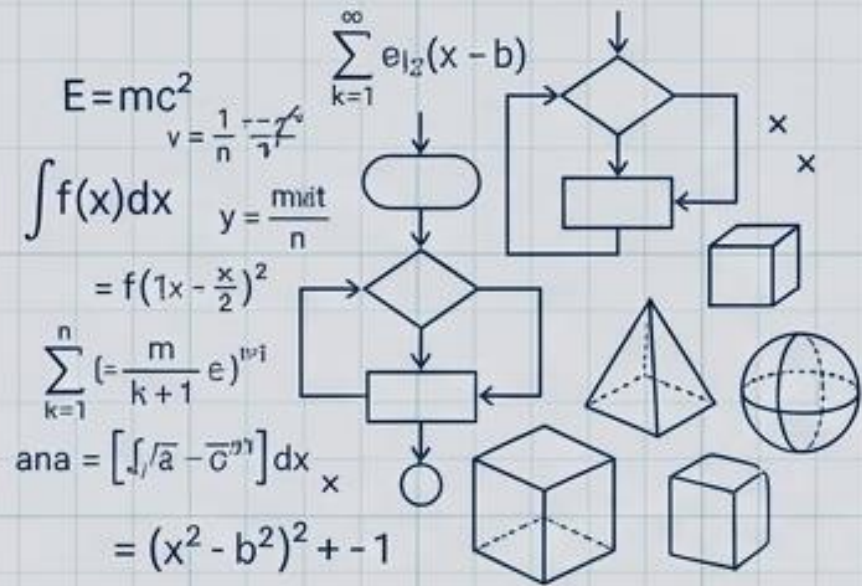
The Universal Default: Copyright Law



The Boundary of Protection: Expression vs. Idea



NOT PROTECTED



Copyright does **not** protect **ideas**, **procedures**, **methods of operation**, or **mathematical concepts**.



COPYRIGHT PROTECTED



Copyright protects the **particular expression** of an idea and **non-functional, artistic elements** (like combined designs or specific code implementations).

The Owner's Control Panel

This bundle allows owners to realize commercial gain by selling or licensing any combination of these rights.

Reproduction

The right to make **direct copies** of a protected work.

Distribution

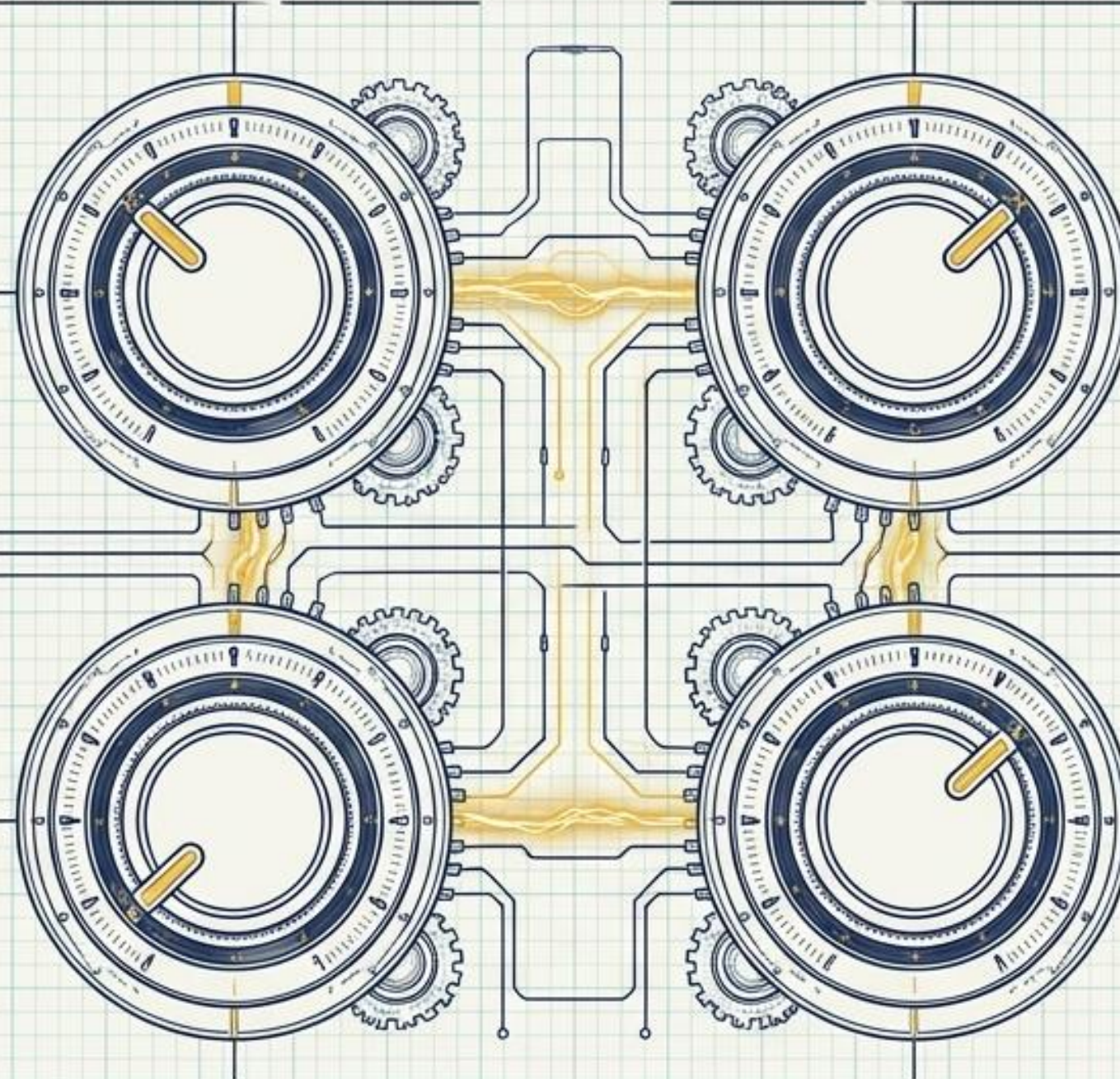
The right to **sell or otherwise distribute** copies to the public.

Adaptation

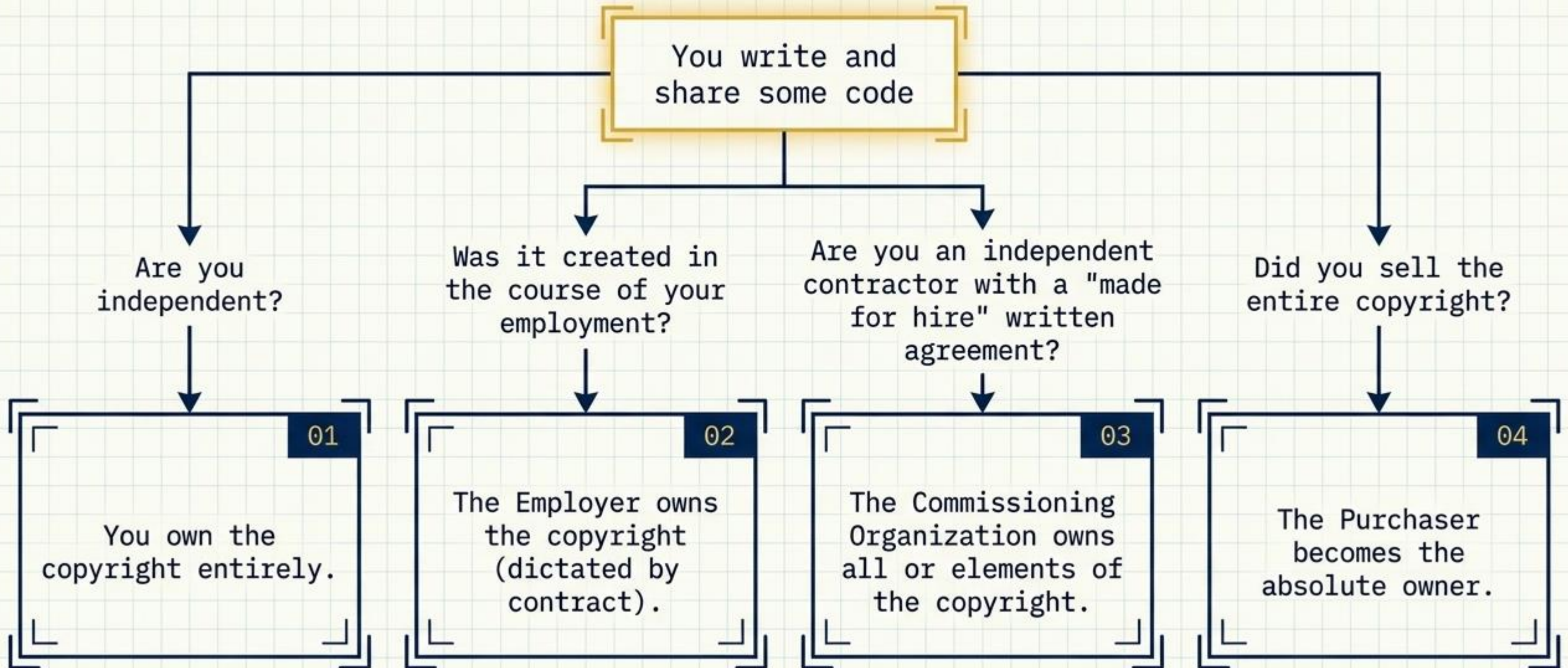
The right to prepare new **works based** on the original (Derivative Works).

Performance

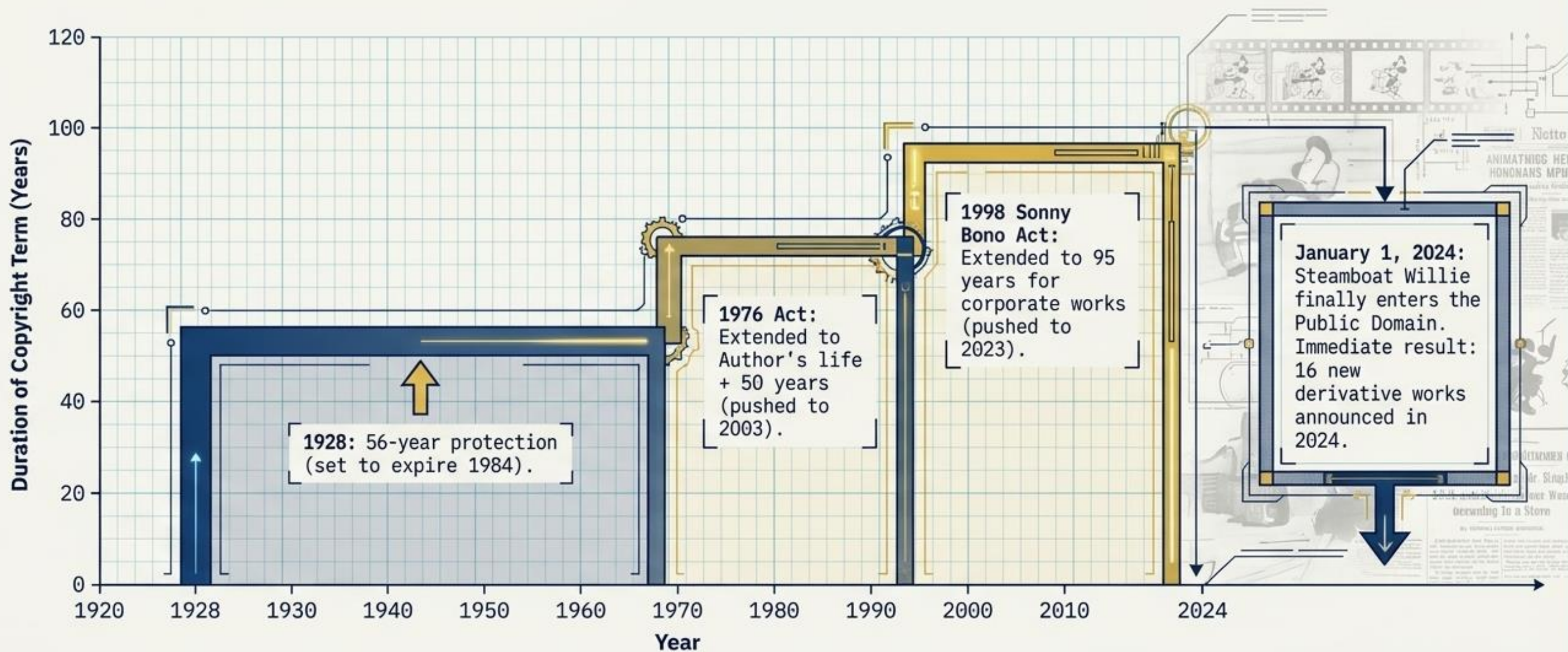
The right to **perform or display** a protected work publicly.



Who Actually Owns the Code?



Glitches in the Matrix: The House of Mouse



Perpetual Anomalies in UK Law

The Hospital Exception (Peter Pan)

Panel 1

J.M. Barrie gifted rights to Great Ormond Street Hospital.



Hack

When copyright expired in 1987, the 1988 CDPA added a special extension allowing the hospital to collect royalties perpetually (though without creative control).

The Crown Exception (King James Bible)

Panel 2

The Copyright Act of 1775.



Hack

Established a perpetual crown copyright for printed editions in the UK, restricting printing solely to the Royal printer and Oxford/Cambridge Universities.

The Fair Dealing Sandbox & The Public Domain

Copyright facilitates relaxations to the law through Fair Use/Dealing, provided they do not affect market opportunities.

Academic/News

Research, private study, criticism, review, and news reporting.

Educational/Official

Royal commissions, statutory enquiries, judicial proceedings, performance/lending for education, libraries.

Personal Utility & Critique

Format shifting/backups (including computer programs), time-shifting broadcasts, parody, pastiche, and incidental inclusion.

The Public Domain



Creative works to which no exclusive rights apply because they have: Expired, Been Forfeited, Been Expressly Waived, or are Inapplicable.

Applying the Law to Code: Software Licences

A legal instrument governing the use or redistribution of software.

Proprietary Software

Mechanism

Uses restrictive copyright licensing.

Source Code

Hidden from the user.

Philosophy

Permissively enables only limited use of the software and artefacts.

Free and Open-Source Software (FOSS)



Mechanism

Anyone is freely licensed to use, copy, study, and change the software.

Source Code

Openly shared to encourage voluntary improvement.

The Philosophy of Free: The Four Freedoms

"Free as in free speech,
not as in free beer."

Modification Freedoms

Freedom 1: The freedom to study how the program works and change it.
Freedom 3: The freedom to distribute copies of your modified versions to benefit the community.

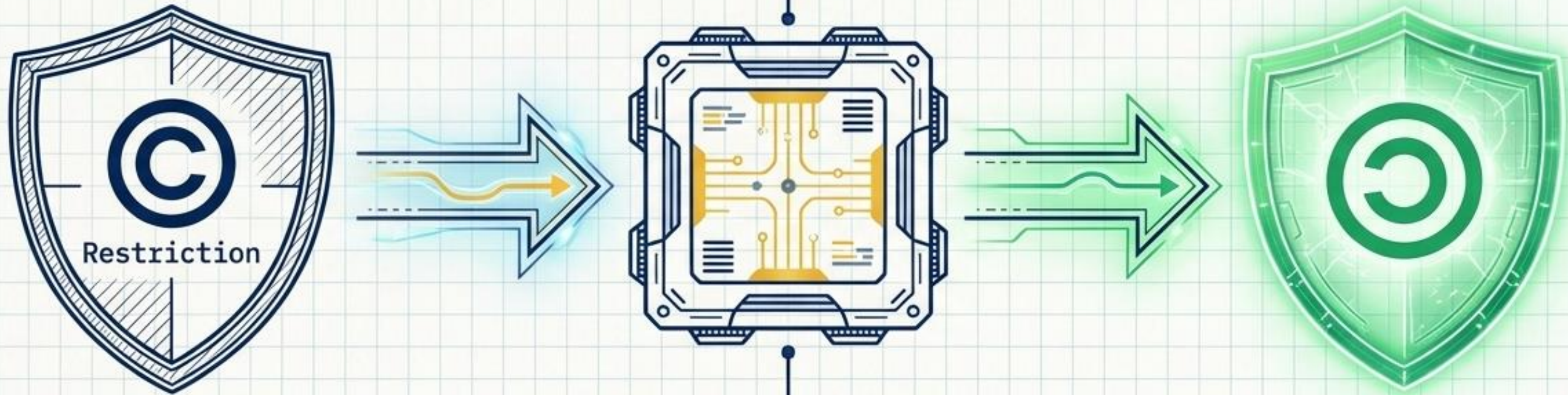
Access to the Source
Code is a Precondition

Binary Freedoms

Freedom 0: The freedom to run the program as you wish, for any purpose.
Freedom 2: The freedom to redistribute copies so you can help others.

The Ingenuity of Copyleft: A Legal Hack

FOSS licences leverage restrictive copyright law to ensure that freedoms are imbued within software legally. This practice is known as Copyleft.



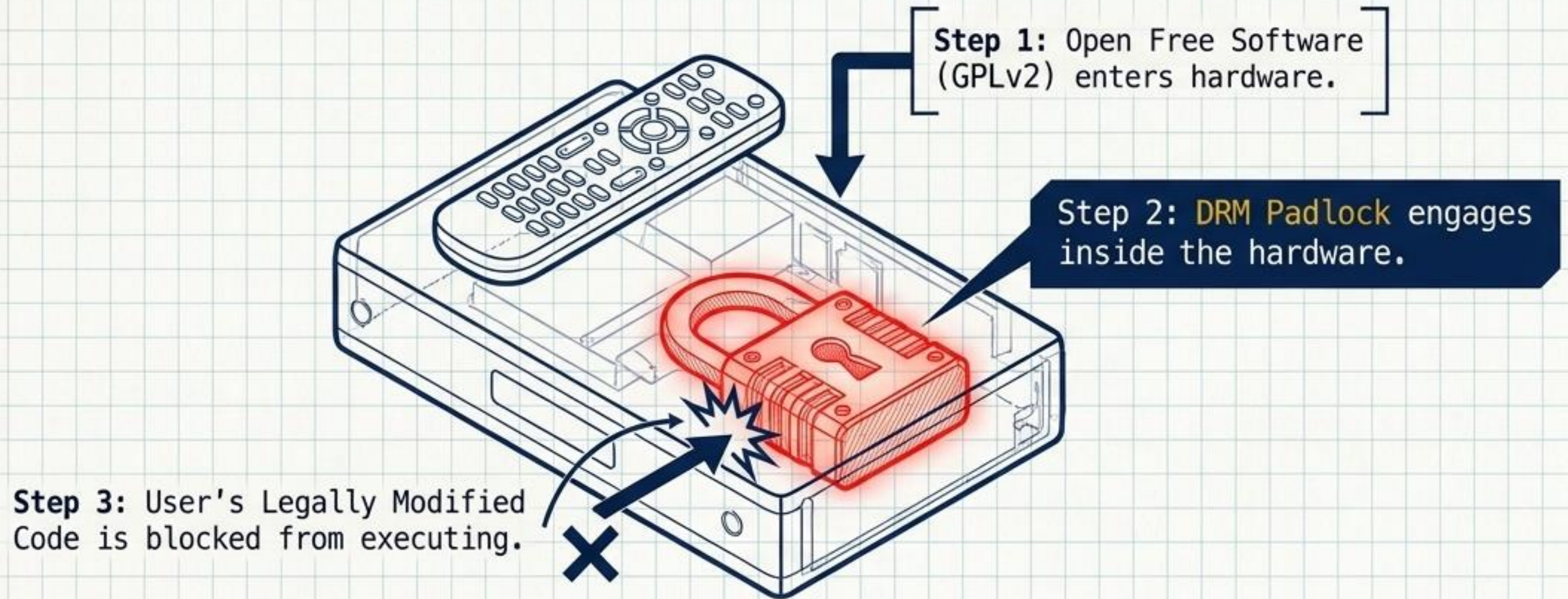
Copyleft grants the right to freely distribute and modify IP with the strict requirement that the exact same rights are preserved in all derivative works. It mandates the sharing of source code and license terms.

Diagnostic Matrix: Default vs. Copyleft

Dimension	Copyright (Default)	Copyleft
Aim	Control / Providing restriction.	Public exchange.
Provided By	Inherent Legal frameworks.	Adopting a specific license (e.g., GPL).
Distribution	Restricted by author's wish.	Allowed freely.
Modification & Redistribution	Generally prohibited.	Generally permissive, relative to the goals of an author (protective/reciprocal).

The Hardware Loophole: Tivoization

The first broadly adopted copyleft license was Richard Stallman's GPL. The Linux Kernel famously uses GPLv2.



Tivoization occurs when a system incorporates copyleft software but uses **hardware** restrictions (**DRM**) to prevent users from running their modified versions. The software is free, but the hardware builds a prison around it. **GPLv3** was authored specifically to address this practice.